

title

Observación 1. Sea $f \in \mathcal{H}(\Omega)$, entonces

$$f'(z) \stackrel{\text{def}}{=} \lim_{\substack{h \rightarrow 0 \\ \text{en } \mathbb{C}}} \frac{f(z+h) - f(z)}{h} = \lim_{\substack{s \rightarrow 0 \\ \text{en } \mathbb{R}}} \frac{f(z+s) - f(z)}{s} = \frac{\partial f}{\partial x}(z) = -i \frac{\partial f}{\partial y}(z).$$